

Evaluation

Inconsistency Checker · v0.8.8 · 2026-08-18 · fixed seed 7

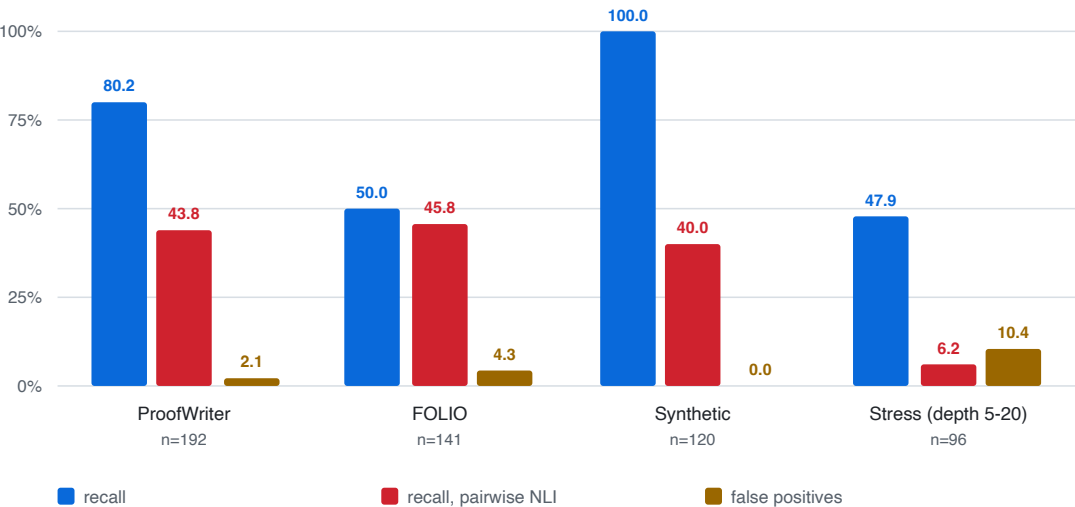
The system extracts claims from a document, translates each into first-order logic, and gives every logical judgement to the Z3 solver. It reports the *minimal* set of statements that cannot all be true, together with a derivation of the conflict.

Evaluated on four held-out datasets, none of which was used during development. Two are externally authored with external labels; two are generated so their labels follow from construction rather than from inspection. Every result is compared against a **sentence-pair entailment (NLI) baseline** run on identical documents under identical scoring.

Headline results

Results by held-out dataset

this system against sentence-pair entailment, identical documents and scoring



Dataset	n	Source	Recall (95% CI)	Precision	False positives
ProofWriter	192	external	80.2% [71.1–87.0]	97.5%	2.1%
FOLIO	141	external	50.0% [38.8–61.3]	92.3%	4.3%
Synthetic	120	constructed	100% [94.0–100]	100%	0.0%
Stress (depth 5–20)	96	constructed	47.9% [34.5–61.7]	82.1%	10.4%

Balanced by construction (ProofWriter 96/96, FOLIO 72/69, synthetic 60/60, stress 48/48), so recall and false-positive rate are measured on equal numbers of positive and negative documents. Intervals are Wilson score intervals.

Recall does not degrade with reasoning depth

Recall vs. reasoning depth

ProofWriter: 192 held-out documents, externally authored



The central claim. A contradiction distributed across k inference steps is invisible to any method that compares sentences pairwise: at $k \geq 2$ **no pair of sentences is inconsistent**, so pairwise entailment cannot see the conflict at all.

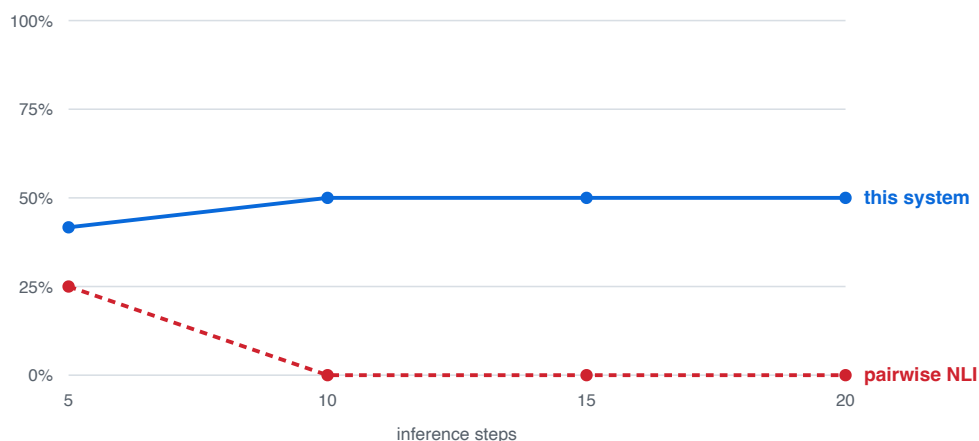
This is measured, not asserted. The NLI baseline scores **92.3% at depth 0**, where the contradiction is directly visible in one pair, then **12.5% at depth 1** and **0% from depth 3 onward**. The full pipeline holds 56–81% through depth 5.

The dashed purple line is the same pipeline with the language model removed from translation (§ ablation). It collapses to 0% beyond depth 1 for a different reason: coverage, not structure.

Beyond the public corpora: 20 inference steps

Deeper chains: 5 to 20 inference steps

purpose-built stress set, 96 documents, up to 100 irrelevant statements



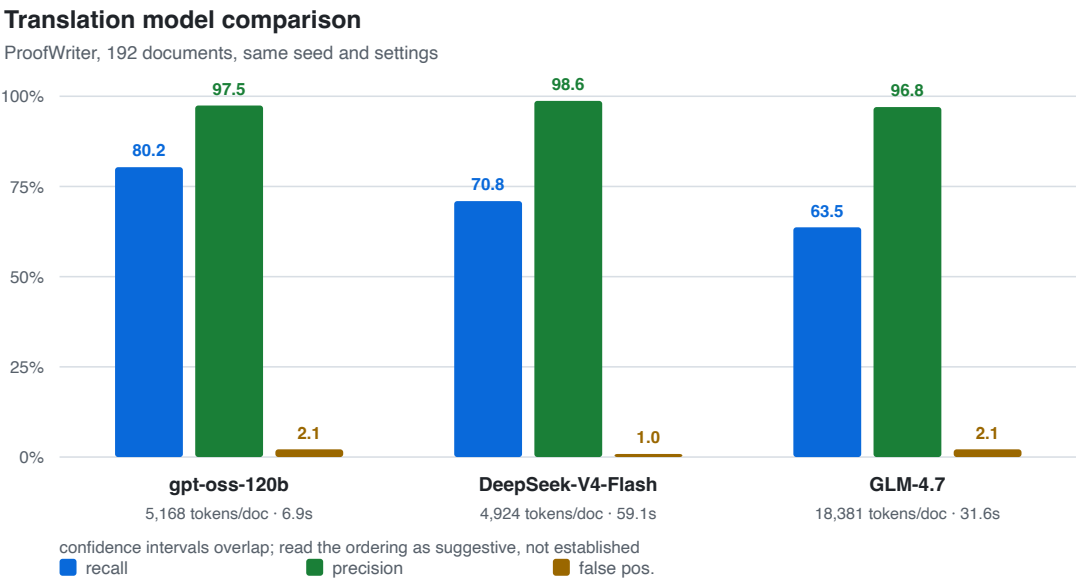
ProofWriter tops out at 5 steps. A purpose-built set extends the axis to 20, crossed with 0, 40 and 100 irrelevant statements so that depth and document length vary independently. Distractors reuse the same predicate vocabulary on *other* entities, so they cannot be dismissed by surface similarity.

Depth 5–20	Recall	False positives	Model calls per document
This system	47.9%	10.4%	~3 + solver
Sentence-pair (NLI)	6.2%	0.0%	127.2

The pairwise baseline reaches 25% at depth 5 and **0% at every deeper level**, while making 127 model calls per document, and its cost grows with the square of document length, so it becomes most expensive exactly where it stops working. Our recall is *flat* across depth, because a solver is indifferent to chain length.

That flatness is also the diagnosis: recall sits near 50% at **every** depth including the shallowest, so the loss is not depth-related. A document fails when any single statement fails to translate, and one missing link breaks a chain of any length.

Model comparison and the LLM's contribution



Model	Recall	Precision	FP	Tokens/doc	s/doc
gpt-oss-120b	80.2%	97.5%	2.1%	5,168	6.9
DeepSeek-V4-Flash	70.8%	98.6%	1.0%	4,924	59.1
GLM-4.7	63.5%	96.8%	2.1%	18,381	31.6
rule-only (LLM withheld)	12.5%	92.3%	1.0%	0	offline

Ablation. Withholding the LLM translator while leaving extraction, the gate, the solver and reporting untouched drops recall from **80.2% → 12.5%** (and 100% → 1.7% on the synthetic set). Recall is therefore bounded by *translation coverage*, not by the reasoning engine: the deterministic translator currently produces usable logic for about 5% of real sentences. This is the primary target for future work.

Precision never falls below 92% in any configuration, including with no LLM at all. The system does not manufacture contradictions.

Model recall differences have overlapping confidence intervals and should be read as suggestive, not established. The models were not run under identical settings: gpt-oss cannot disable internal reasoning on its provider, DeepSeek can.

Method

Every verdict is a Z3 proof, not a model judgement. The language model only proposes formulas. Content outside the supported fragment (modality, tense, arithmetic, comparatives, hedged generalisations) is **quarantined with a written reason** rather than guessed at; quarantine rates were 3.2% (ProofWriter), 7.5% (FOLIO), 2.7% (synthetic). ProofWriter and FOLIO are entailment datasets, converted to consistency checking by

appending the negated conclusion: a theory entails Q exactly when the theory together with $\neg Q$ is inconsistent. The conversion rule was fixed before any result was seen.

Baseline. Sentence-pair entailment judges every pair of statements independently and flags the document if any pair conflicts. Each judgement sees only its two sentences. That isolation is the method, so pairs are never batched into one prompt, which would let the model reason across them. It is scored by the same rule and the same labels as the pipeline. On the two external corpora it reaches 43.8% and 45.8% recall, at 7.1% and 18.8% false positives.

Reproducibility. Fixed seed and temperature 0; the full test suite (255 tests) runs offline against shipped fixtures with no API key. Datasets, adapters, scorer, baseline and charts are in `tools/`; per-document outputs and `metrics.json` for every run are retained.

Limitations. The generated sets were authored by us and exercise the fragment the system targets; the external results are the load-bearing ones. The depth-3 dip is non-monotonic across all three models and is a sampling artefact at 16 documents per cell, not a depth effect. Performance tracks how *logical* a contradiction is: conflicts resting on arithmetic, dates, degree or world knowledge are outside the representation by design. The comparison here is against sentence-pair entailment specifically; prompting a language model to judge a whole document at once is a different approach that does not face the pairwise structural limit, and is not evaluated in this report.